

# A Counterexample to Cell-Average Positivity for Two-Dimensional Backward Euler DG Methods

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October 1, 2026

## Abstract

We give an explicit counterexample for the standard, exactly integrated, upwind discontinuous Galerkin discretization of a constant-coefficient scalar advection equation with backward Euler time stepping on a Cartesian mesh. For each of the fixed approximation spaces  $P^1$  and  $Q^1$ , the same mesh and the same pointwise nonnegative piecewise affine initial condition produce a strictly negative average in the same cell after one time step, for every positive time step. The proof gives exact rational formulas for the offending averages. No source term, negative inflow data, or time-step-dependent initial condition is used.

## 1 Background and problem formulation

Implicit positivity-preserving DG methods with backward Euler time discretization were developed by Qin and Shu [3]. Ling, Cheng, and Shu [1] proved cell-average positivity for a one-dimensional upwind transport solver and exhibited its failure for standard two-dimensional  $P^1$  and  $Q^1$  spaces; see their Section 4.1 and Example 5. Their explicit two-dimensional examples use nonnegative source terms in steady transport problems. In particular, those examples do not directly supply a fixed, nonnegative degree-one initial polynomial for the source-free transient problem considered here. Milani, Renac, and Ruel [2] analyzed backward Euler DGSEM on Cartesian grids, proved a one-dimensional cell-average maximum principle for sufficiently large time steps, and used flux-corrected transport with graph viscosity to enforce it in multiple dimensions.

The present note gives a self-contained algebraic construction for the exactly integrated DG method: one fixed mesh and one fixed nonnegative initial polynomial yield a negative cell average for every positive time step in each of the fixed spaces  $P^1$  and  $Q^1$ . We make no claim that this note exhausts prior counterexamples. The distinction between exactly integrated DG and quadrature-based DGSEM is retained throughout.

Consider

$$u_t + u_x + u_y = 0 \quad \text{in } \Omega = (0, 5) \times (0, 3), \quad (1.1)$$

with homogeneous inflow data on  $x = 0$  and  $y = 0$  and natural outflow boundaries on  $x = 5$  and  $y = 3$ . Let

$$K_{ij} = (i - 1, i) \times (j - 1, j), \quad 1 \leq i \leq 5, \quad 1 \leq j \leq 3,$$

and let  $\mathcal{T}_h$  consist of these fifteen unit squares. On each cell, choose either

$$P^1(K) = \text{span}\{1, \xi, \eta\} \quad \text{or} \quad Q^1(K) = \text{span}\{1, \xi, \eta, \xi\eta\},$$

where  $\xi = 2(x - i + \frac{1}{2})$  and  $\eta = 2(y - j + \frac{1}{2})$  are the reference coordinates on  $K_{ij}$ . The choice is uniform over all cells and fixed independently of the time step. Denote the corresponding discontinuous finite element space by  $V_h$ .

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Set  $\beta = (1, 1)^T$  and  $\lambda = \Delta t > 0$ , since both mesh widths equal one. Each directional Courant number is  $\lambda$ ; the sum of the directional Courant numbers is  $2\lambda$ . Given  $u_h^0 \in V_h$ , the unlimited backward Euler DG update seeks  $u_h^1 \in V_h$  such that, for every  $K \in \mathcal{T}_h$  and every  $v \in V_h|_K$ ,

$$\int_K (u_h^1 - u_h^0) v \, dx \, dy - \lambda \int_K u_h^1 (v_x + v_y) \, dx \, dy + \lambda \int_{\partial K} (\beta \cdot \mathbf{n}_K) \widehat{u}_h^1 v \, ds = 0. \quad (1.2)$$

All integrals are exact. At an interior face,  $\widehat{u}_h^1$  is the trace from the upwind cell. At a physical inflow face it is zero; at a physical outflow face it is the interior trace. Thus no limiter, mass lumping, graph viscosity, or flux correction is included in (1.2). The cell average is

$$\bar{u}_{ij}^1 = \frac{1}{|K_{ij}|} \int_{K_{ij}} u_h^1 \, dx \, dy.$$

## 2 A fixed counterexample for every positive Courant number

**Theorem 2.1** (Failure of cell-average positivity for all positive time steps). *For either  $V_h|_K = P^1(K)$  or  $V_h|_K = Q^1(K)$ , take*

$$u_h^0(x, y) = \begin{cases} 2(y - 1), & (x, y) \in K_{22}, \\ 0, & (x, y) \in K, \quad K \neq K_{22}. \end{cases} \quad (2.1)$$

*Every cellwise polynomial is nonnegative on its closed cell, and  $0 \leq u_h^0 \leq 2$ . For every  $\lambda > 0$ , the update (1.2) has a unique solution and*

$$\bar{u}_{42}^1 < 0.$$

*More precisely, define*

$$p(\lambda) = 1 + 6\lambda + 14\lambda^2, \quad (2.2)$$

$$q(\lambda) = 1 + 8\lambda + 24\lambda^2, \quad (2.3)$$

$$N_P(\lambda) = 1 + 20\lambda + 142\lambda^2 + 438\lambda^3 + 612\lambda^4 + 396\lambda^5, \quad (2.4)$$

$$N_Q(\lambda) = 1 + 27\lambda + 282\lambda^2 + 1386\lambda^3 + 3312\lambda^4 + 5616\lambda^5 + 10368\lambda^6. \quad (2.5)$$

*Then the averages are*

$$\bar{u}_{42}^1 = -\frac{2\lambda^2 N_P(\lambda)}{(1 + 4\lambda)^2 p(\lambda)^3} \quad \text{for } P^1, \quad (2.6)$$

*and*

$$\bar{u}_{42}^1 = -\frac{2\lambda^2 N_Q(\lambda)}{(1 + 4\lambda)^3 q(\lambda)^3} \quad \text{for } Q^1. \quad (2.7)$$

We first record the local algebra needed in the proof. Coefficient vectors below are column vectors in the displayed reference bases.

**Lemma 2.1** (Local systems with zero bottom inflow). *Suppose that a cell has zero bottom inflow. Let  $F$  be the coefficient vector of its initial polynomial, and let  $U_L$  be the coefficient vector of the new-time polynomial in its left neighbor (or the zero vector at a homogeneous left inflow boundary). The local equation is*

$$HU = F + BU_L. \quad (2.8)$$

For  $P^1$ , the matrices are

$$H_P = \begin{pmatrix} 1+2\lambda & \lambda & \lambda \\ -3\lambda & 1+4\lambda & 0 \\ -3\lambda & 0 & 1+4\lambda \end{pmatrix}, \quad B_P = \lambda \begin{pmatrix} 1 & 1 & 0 \\ -3 & -3 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (2.9)$$

For  $Q^1$ , they are

$$H_Q = \begin{pmatrix} 1+2\lambda & \lambda & \lambda & 0 \\ -3\lambda & 1+4\lambda & 0 & \lambda \\ -3\lambda & 0 & 1+4\lambda & \lambda \\ 0 & -3\lambda & -3\lambda & 1+6\lambda \end{pmatrix}, \quad B_Q = \lambda \begin{pmatrix} 1 & 1 & 0 & 0 \\ -3 & -3 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & -3 & -3 \end{pmatrix}. \quad (2.10)$$

The local diagonal matrices, also when the bottom inflow is nonzero, are invertible for every  $\lambda > 0$ , with

$$\det H_P = (1+4\lambda)p(\lambda), \quad \det H_Q = (1+4\lambda)^2q(\lambda). \quad (2.11)$$

*Proof.* For reference polynomials, write

$$\langle f \rangle = \frac{1}{4} \int_{-1}^1 \int_{-1}^1 f(\xi, \eta) d\xi d\eta, \quad \langle g \rangle_\eta = \frac{1}{2} \int_{-1}^1 g(\eta) d\eta, \quad \langle g \rangle_\xi = \frac{1}{2} \int_{-1}^1 g(\xi) d\xi.$$

If  $u$  is the unknown polynomial and  $\ell(\eta) = u_L(1, \eta)$  is its left inflow trace, (1.2) becomes

$$\begin{aligned} & \langle uv \rangle - 2\lambda \langle u(v_\xi + v_\eta) \rangle + \lambda \langle u(1, \eta)v(1, \eta) \rangle_\eta + \lambda \langle u(\xi, 1)v(\xi, 1) \rangle_\xi \\ & = \langle u_h^0 v \rangle + \lambda \langle \ell(\eta)v(-1, \eta) \rangle_\eta. \end{aligned} \quad (2.12)$$

The factor 2 in the volume term follows from  $v_x = 2v_\xi$  and  $v_y = 2v_\eta$ .

For  $P^1$ , take  $v = 1, \xi, \eta$  and use  $\langle \xi \rangle = \langle \eta \rangle = 0$  and  $\langle \xi^2 \rangle = \langle \eta^2 \rangle = 1/3$ . The reference mass matrix is  $M_P = \text{diag}(1, 1/3, 1/3)$ . Premultiplying the weak equations by  $M_P^{-1}$  gives (2.9). For  $Q^1$ , additionally take  $v = \xi\eta$ , use  $\langle \xi^2 \eta^2 \rangle = 1/9$ , and premultiply by  $M_Q^{-1}$ , where  $M_Q = \text{diag}(1, 1/3, 1/3, 1/9)$ . This gives (2.10). The initial-data term is  $F$  after this normalization. A nonzero bottom inflow adds only a known right-hand-side term and does not change  $H$ . Expanding the determinants gives (2.11); their factors are positive for  $\lambda > 0$ .  $\square$

*Proof of Theorem 2.1.* The initial polynomial in  $K_{22}$  is  $1 + \eta$ , so (2.1) belongs to both spaces and is pointwise nonnegative on every cell.

For the velocity  $(1, 1)$ , each local equation couples its unknown only to the new-time polynomials in the left and bottom neighbors. Ordering cells by increasing  $i + j$  makes the global system block lower triangular. Its diagonal blocks are  $H_P$  or  $H_Q$ , all invertible by Lemma 2.1. Hence the full update has a unique solution.

All cells in the bottom row have zero initial data and homogeneous bottom inflow. Sweeping from left to right shows that their new-time solutions are zero. Likewise, the new-time solution in  $K_{12}$  is zero. Consequently, the bottom inflow into  $K_{22}, K_{32}, K_{42}$  and the left inflow into  $K_{22}$  are zero. The solutions in cells above this row or to the right of  $K_{42}$  cannot affect these three cells.

Let  $U_0, U_1, U_2$  denote the coefficient vectors in  $K_{22}, K_{32}, K_{42}$ , respectively. In the  $P^1$  case put  $F_P = (1, 0, 1)^T$ , and in the  $Q^1$  case put  $F_Q = (1, 0, 1, 0)^T$ . The local systems are exactly

$$HU_0 = F, \quad HU_1 = BU_0, \quad HU_2 = BU_1. \quad (2.13)$$

Since all nonconstant basis functions have zero cell average,  $\bar{u}_{42}^1 = e_0^T U_2$ , where  $e_0$  is the first coordinate vector. Thus

$$\bar{u}_{42}^1 = e_0^T H^{-1} B H^{-1} B H^{-1} F. \quad (2.14)$$

For completeness, the exact polynomial identities that evaluate this matrix product are

$$e_0^T \operatorname{adj}(H_P) B_P \operatorname{adj}(H_P) B_P \operatorname{adj}(H_P) F_P = -2\lambda^2(1 + 4\lambda)N_P(\lambda), \quad (2.15)$$

$$e_0^T \operatorname{adj}(H_Q) B_Q \operatorname{adj}(H_Q) B_Q \operatorname{adj}(H_Q) F_Q = -2\lambda^2(1 + 4\lambda)^3 N_Q(\lambda). \quad (2.16)$$

Here  $\operatorname{adj}$  denotes the classical adjugate. These are polynomial identities, obtained by multiplying the matrices in (2.9) and (2.10) and collecting powers of  $\lambda$ . Using  $H^{-1} = \operatorname{adj}(H)/\det H$ , substituting (2.15)–(2.16) and (2.11) into (2.14), and canceling the positive powers of  $1 + 4\lambda$  gives (2.6) and (2.7).

Every coefficient of  $p, q, N_P, N_Q$  is positive. Since  $\lambda > 0$ , both displayed averages are strictly negative. The mesh, the approximation degree, the initial condition, and the offending cell are the same for every positive time step.  $\square$

**Corollary 2.1** (Neither an upper nor a lower CFL bound suffices). *For either fixed space  $P^1$  or  $Q^1$ , no restriction of the form  $0 < \lambda \leq C$  or  $\lambda \geq C$ , with  $C > 0$ , guarantees nonnegative cell averages after a backward Euler DG step for all pointwise nonnegative initial polynomials. The same statement holds when CFL is defined as  $2\lambda$ .*

*Proof.* The data in Theorem 2.1 give a negative average for every  $\lambda > 0$ , including every time step in either specified range.  $\square$

*Remark 2.1* (Scope and small-step behavior). The theorem concerns the unlimited linear update (1.2) with exact integration. It does not assert the same result for a nonlinear scheme with a limiter embedded in its implicit equation, or for a different quadrature-based DGSEM. Also, membership of the initial data in  $P^k$  or  $Q^k$  for  $k > 1$  does not establish a counterexample for those larger spaces: their discrete operators differ. The degree in this theorem is fixed at  $k = 1$ .

For both spaces, (2.6)–(2.7) yield

$$\bar{u}_{42}^1 = -2\lambda^2 + \mathcal{O}(\lambda^3) \quad (\lambda \downarrow 0).$$

Thus arbitrarily small positive steps still produce a negative average, although its magnitude tends to zero.

## References

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